

Deep Spatial Guidance Framework with Proxemic Interaction Field for Socially Compliant Robot Cruising in Dynamic Scenarios

Supplementary Material

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This supplementary material contains the SN-1 questionnaire, volunteer demographic information, a literature summary table, as well as additional figures and tables from the simulation and static-scenario experiments.

I. ADDITIONAL TABLES

TABLE S.I
SN-1 SCALE: SUBJECTIVE EVALUATION QUESTIONNAIRE FOR ROBOT SOCIAL CRUISING IN DYNAMIC SCENARIOS

Date: _____ **Participant ID:** _____ **Scenario:** _____

Instructions: Based on your observation of and interaction with the robot, please rate your level of agreement with each statement below. Mark (✓) the box that best reflects your opinion.

Scale: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

No.	Statement	Score				
		1	2	3	4	5
1	The robot maintained a safe physical distance from pedestrians throughout the interaction.	☐	☐	☐	☐	☐
2	The robot effectively avoided collisions with dynamic obstacles and pedestrians.	☐	☐	☐	☐	☐
3	The robot respected social groups (e.g., interacting people) and did not intrude into their interaction space.	☐	☐	☐	☐	☐
4	The robot's avoidance behavior and trajectory appeared natural and socially acceptable.	☐	☐	☐	☐	☐
5	The robot's motion was smooth, without unnecessary oscillations or sudden stops.	☐	☐	☐	☐	☐
6	The robot navigated the environment efficiently without showing hesitation in crowded areas.	☐	☐	☐	☐	☐
7	I felt psychologically comfortable and relaxed sharing the space with the robot.	☐	☐	☐	☐	☐
8	I would accept this robot operating in real-world public environments (e.g., malls, airports).	☐	☐	☐	☐	☐

Additional Comments (Optional):

Thank you for your participation!

TABLE S.II
DEMOGRAPHIC INFORMATION OF VOLUNTEERS

ID	Static	Dynamic	Age	Gender	Major	Previous Contacts
1	1, 2	1, 2	22	M	Automation	None
2	1, 2	1, 2	25	M	Electronic Engineering	None
3	1, 2	1	20	F	Computer Science	None
4	1, 2	1, 2	24	M	Automation	2
5	1, 2	1, 2	27	M	Mechanical Engineering	None
6	1, 2	—	18	F	Automation	None
7	1, 2	—	23	M	Computer Science	1
8	1, 2	2	26	M	Automation	None
9	1, 2	—	21	M	Automation	1
10	1, 2	—	27	M	Electronic Engineering	None
11	1, 2	1	24	M	Automation	2
12	1, 2	—	20	M	Computer Science	None
13	1, 2	2	26	M	Automation	None
14	—	1, 2	19	F	Mechanical Engineering	None
15	—	1, 2	23	M	Automation	2
16	—	2	25	F	Automation	None
17	—	1, 2	26	M	Automation	1
18	—	1	22	M	Computer Science	None
19	—	2	21	F	Automation	None
20	—	2	27	M	Electronic Engineering	None
21	—	1, 2	24	M	Automation	None
22	—	2	25	M	Mechanical Engineering	1

Note: A total of 22 volunteers participated in the practical experiments. Static 1 and Static 2 involved the same group of 13 volunteers (1–13). Dynamic 1 involved 11 volunteers; Dynamic 2 involved 14 volunteers, some volunteers participated in both Dynamic 1 and Dynamic 2. Four volunteers (1, 2, 4, and 5) participated in all four experiments. The volunteers were aged 18–27 and comprised 17 males and 5 females. The majority were Automation majors, with the remainder from Computer Science, Electronic Engineering, and Mechanical Engineering. Fifteen volunteers had no previous contacts with robots; four had 1 previous contact; three had 2 previous contacts.

TABLE S.III
STATISTICAL COMPARISON OF SUBJECTIVE EVALUATION SCORES IN STATIC SCENARIOS

Method	Static 1	Static 2	Overall (Total)
Standard RRT* [25]	3.29 ± 0.72	3.40 ± 0.51	3.34 ± 0.61
Informed RRT* [26]	3.45 ± 0.41	3.83 ± 0.46	3.64 ± 0.47
Neural RRT* [27]	4.04 ± 0.37	3.94 ± 0.65	3.99 ± 0.52
DSG-RRT* (Ours)	4.20 ± 0.39	4.16 ± 0.34	4.18 ± 0.35

* The values are presented as Mean ± Standard Deviation (SD).

* Scale scores range from 0 to 5, higher values indicate better evaluation.

TABLE S.IV
VALIDITY AND RELIABILITY ANALYSIS OF THE SN-1 SCALE IN STATIC SCENARIOS

Metric	Value
KMO Measure of Sampling Adequacy	0.830
Bartlett's Test of Sphericity	
Approx. Chi-Square (χ^2)	164.87
Degrees of Freedom (df)	28
Significance (p)	< 0.001
Cronbach's α Coefficient	0.812

TABLE S.V
SUMMARY OF LITERATURE SURVEYED IN THE INTRODUCTION AND RELATED WORK

Topic	References	Limitation
Socially aware navigation	[1]–[4], [9]–[18]	Focus on guiding robots to avoid pedestrians, paying less attention to proactively approaching or interacting with human groups; tend to encode social compliance as a soft cost that can be overridden when it conflicts with path efficiency.
Group recognition	[19]–[21]	Most rely on a single social cue and treat recognition as a preprocessing step, separating it from planning.
Path planning	[5]–[8], [22]–[31]	Most treat all obstacles uniformly and are not designed to incorporate complex social constraints.

II. ADDITIONAL FIGURES

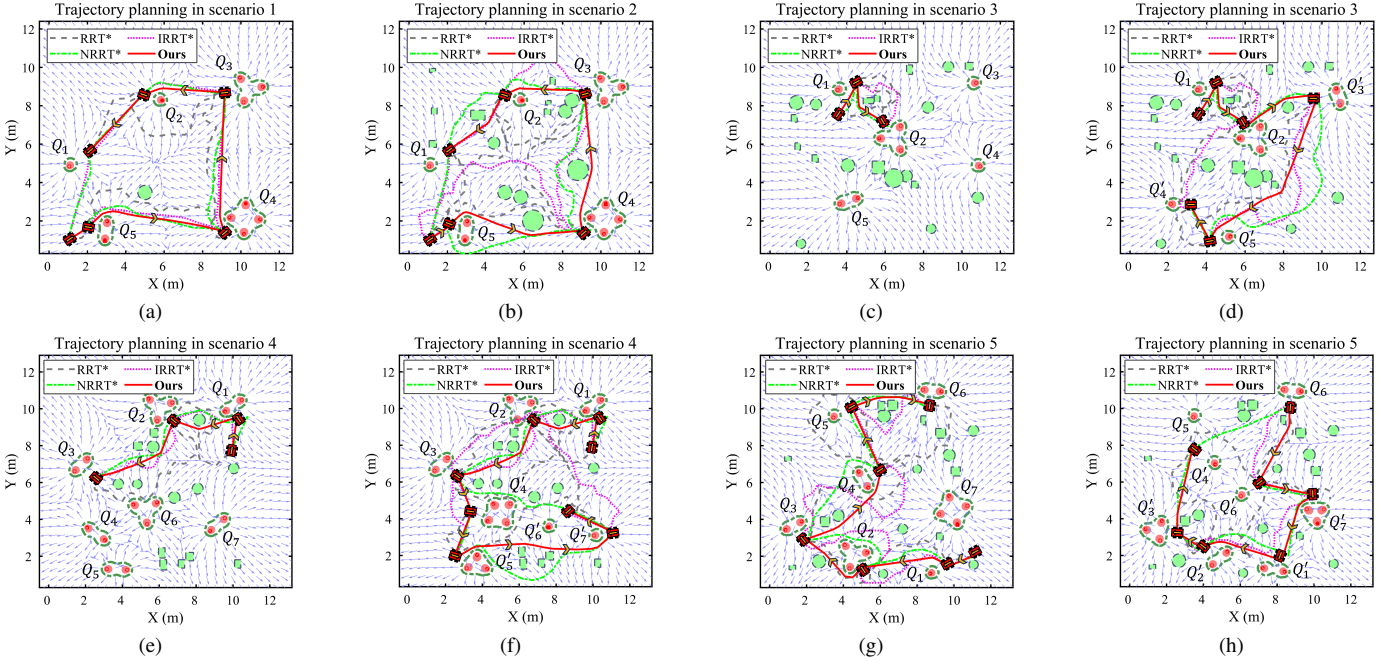


Fig. S.1. Cruising trajectories generated by RRT*, IRRT*, NRRT*, and DSG-RRT* across five randomly generated simulation scenarios. Scenarios 1 and 2 are static; Scenarios 3–5 are dynamic.

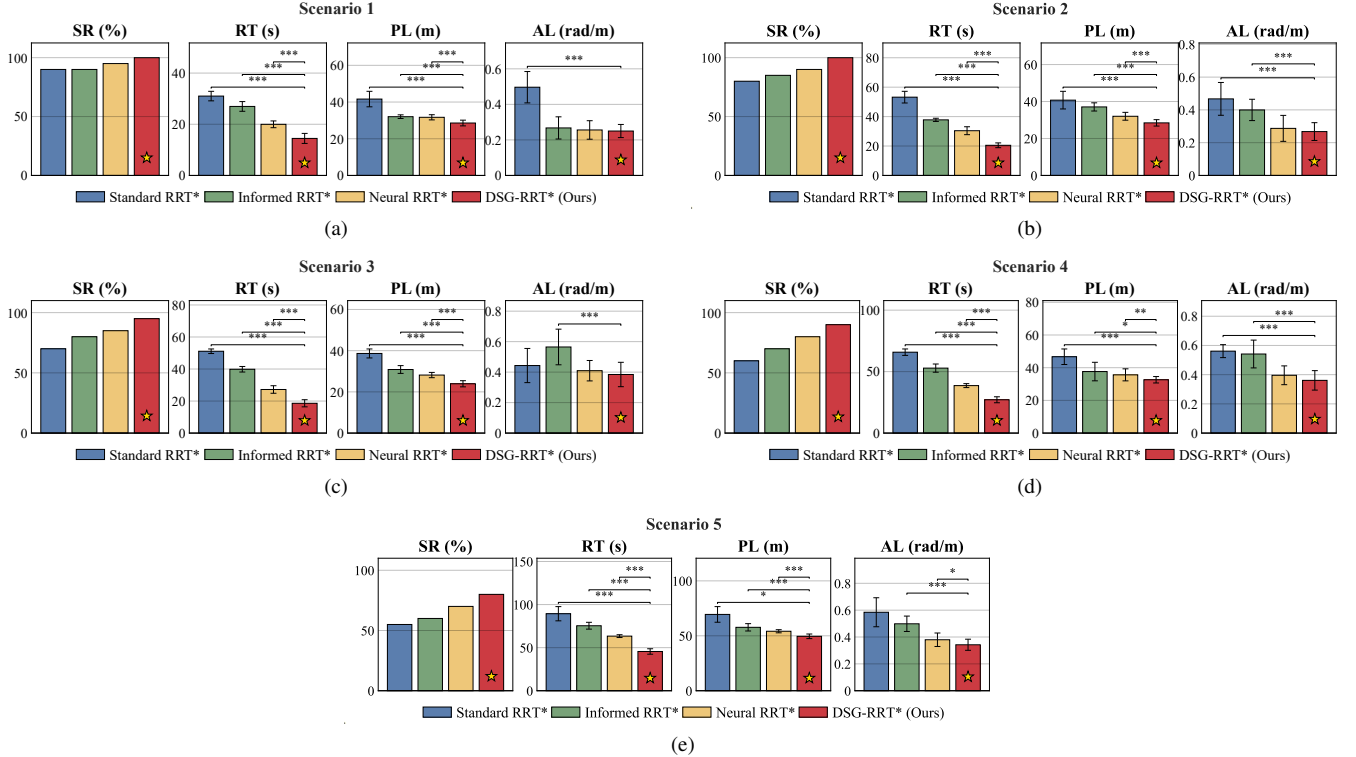


Fig. S.2. Statistical performance evaluation of the comparative methods across simulation scenarios. Efficiency metrics: Success Rate (SR), Running Time (RT), and Path Length (PL). Smoothness metric: Angle over Length (AL). The bars represent mean values, error bars indicate standard deviations, and asterisks denote statistical significance levels. (a) Scenario 1; (b) Scenario 2; (c) Scenario 3; (d) Scenario 4; (e) Scenario 5.

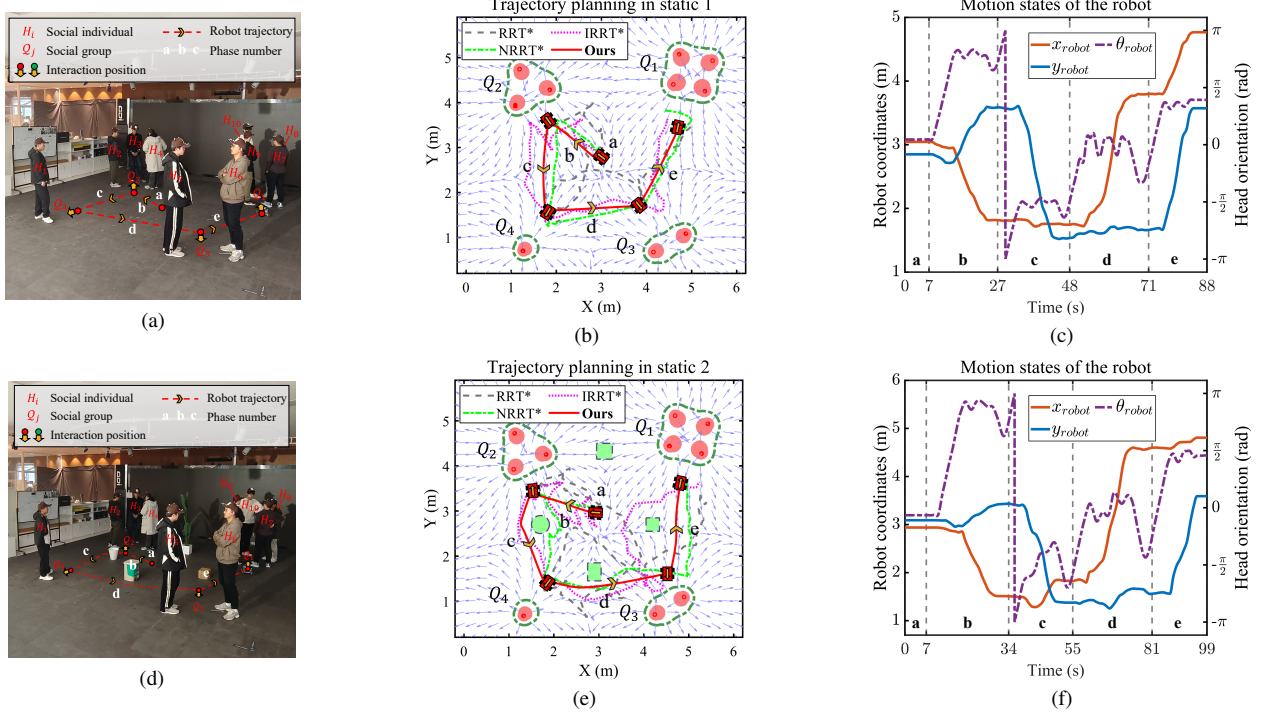


Fig. S.3. Experimental scenarios, cruising trajectories, and motion states in the two static experiments.

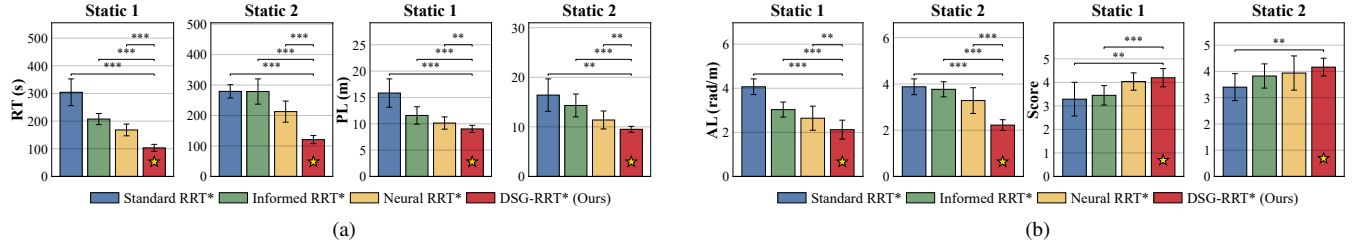


Fig. S.4. Statistical performance evaluation of the comparative methods across static scenarios. (a) Efficiency metrics: Running Time (RT) and Path Length (PL). (b) Smoothness and social compliance metrics: Angle over Length (AL) and Subjective Scores. The bars represent mean values, error bars indicate standard deviations, and asterisks denote statistical significance levels (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).